Universidade Vigo

Evolutionary Computation – Lab-Session 3

Escola Superior de Enxeñaría Informática
Edificio Politécnico
Campus universitario
32004 Ourense

http://esei.uvigo.es
mailto:formella@uvigo.es

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We have fixed the due dates for the first homework as announced already:

- **February 29th** for the minimization of the real-valued functions and the TSP problem applying genetic algorithms.

1. Third Week

Objectives: Dig somewhat deeper into the Guofei-package. Run the genetic algorithm in a Monte Carlo loop to obtain some statistical data. Start to prepare the deliverable (last topic in this sheet!).

1. Run the jupyter notebook on sorting to repeat and to settle what has been presented in the lecture.
2. Run the examples of the Guofei-packages that minimize the Schaffer-function (with both encodings) in a Monte Carlo loop and log the mininum, mean, and variance of the objective function.
3. Try to find a parameter set of the genetic algorithms that you consider sufficiently good. Argue why you took that decision.
4. Replace the Schaffer-function by the Rosenbrock-function (in three or four dimensions) in your code and study what happens: for instance, does the same parameter set, give equally good results? does the search area (defined by the lower and upper bound) have some influence?

```
def rosenbrock(p):
  sum = 0;
  for i in range(len(p)-1):
    xi = p[i];
    xii = p[i+1];
    part = 100*(xii-xi**2)**2 + (xi-1)**2;
    sum+=part;
  return sum
```

5. Use the Guofei-package to find the optimal tour in a Monte Carlo loop and log the mininum, mean, and variance of the objective function. Do you win against the other heuristics given in the jupyter notebook?
6. Feel free to implement one of the mentioned but missing heuristics for the traveling salesperson problem (e.g. farthest insertion) or try yourself with generative AI the same way I mentioned my experiments in class. There is some publically available python code around: https://github.com/afourmy/pyTSP/blob/master/README.md
7. Are the parameters you chose to win against the heuristics on one problem instance well suited over all problems instances? Argue with data.

8. Start to prepare two notebooks: one for the real-valued functions Griewank and Rosenbrock for 2, 3, and 4 dimensions and gather and comment your results regarding the obtained statistical data; one for the TSP problems as given with up to 100 points, again gather and comment your results regarding the obtained statistical data. Add some arguments why you used the settings of the free parameters.